2025_Hexaware_Actual_Test_2

Test Summary

- No. of Sections: 2
- No. of Questions: 90
- Total Duration: 90 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

Additional Instructions:

None

Q1. Find the number of zeros in 135!

- 25
- 30
- 33
- 32

Q2. The expression $px^3 - x^2 + qx + 6$ is divisible by $x + 2$ and has a remainder 30 when divided by $x - 3$. The value of p and q is

- p = −2, q = 7
- p = 7, q = 2
- p = 2, q = 7
- p = 2, q = −7

Q3. 75% of a number exceeds 65% of that number by 80. What is twice that number?

- 800
- 1000
- 1200
- 1600

Q4. A box contains 7 red, 6 white, and 4 blue balls. 3 balls are drawn. If there is one ball of each color, then the number of selections is:

148

124

168

186

Q5. A forgetful teacher had the test papers and the marksheets of 5 students. But, he entered someone else's marks for each of the 5 students.

In how many different ways could he have made this error?

36

44

84

76

Q6. A sum of money lent at compound interest for 2 years at 20% per annum would fetch Rs.723 more if the interest was payable half yearly than if it was payable annually. The sum is

Rs. 20000

Rs. 15000

Rs. 30000

Rs. 45000

Q7. If an amount of Rs.12000 is invested in two different schemes of which one is SI with 10% p.a interest and the other is CI with 10% p.a interest, compounded half-yearly, what will be the difference between SI and CI at the end of 1 year?

50

60

30

40

Q8. Adams and Biden can run a campaign in 8 days, while Carter and Dwight can do the same work in 24 days. In how many days will Adams, Biden, Carter, and Dwight do it together?

12 days

4 days

3 days

6 days

Q9. A 270 metres long train running at the speed of 120 kmph crosses another train running in the opposite direction at the speed of 80 kmph in 9 seconds. What is the length of the other train?

200 m

220 m

240 m

230 m

Q10. A man adds 3 liters of water to 12 liters of milk and another 4 liters of water to 10 liters of milk. What is the ratio of strengths of the milk in the two mixtures?

5/18

24/5

28/25

15/16

Q11. Two vessels contain milk and water in the ratio 3:2 and 7:3. Find the ratio in which the contents of the two vessels to be mixed to get a new mixture in which the ratio of milk and water is 2:1.

2 : 1

1 : 2

4 : 1

1 : 4

Q12. From a book containing 100 pages, one page is selected randomly. What is the probability that the sum of the digits of the page number of the selected page is 11?

11/100

11/50

1/25

2/25

Q13. After selling a fan at a 6% gain and a fridge at a 9% gain, a shopkeeper gains Rs 5100. But if he sells the fan at a 9% gain and the fridge at a 6% loss, he gains Rs 1800 on the whole transaction. Find the original price of the fan.

30000

32000

36000

40000

Q14. A shopkeeper declared a 20% discount on an item. Kanahaiya bought the item from the shop for Rs. 9600 after getting a discount. Kanahaiya sells the item to another person in such a way that he earns a profit of 18% on the marked price. What is the selling price for the second person?

Rs 14000

Rs 12000

Rs 14160

Rs 11460

Q15. The average of 8 numbers is 19. The average of the first and sixth numbers is 17 and that of the third, fifth, and seventh numbers is 24. If the second number is less than the fourth number by 8, and greater than the eighth number by 4, then the sum of a fourth number and the eighth number is:

45

21

13

32

Q16. The area of a parallelogram ABCD is A sq cm. If the distance between AB and DC is d_1 cm and the distance between BC and AD is d_2 cm, then the perimeter of the parallelogram is

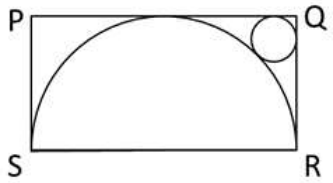
$\frac{2A(d_1 + d_2)}{d_1 d_2}$

$\frac{2A(d_1 + d_2)}{d_1 + d_2}$

$\frac{Ad_1 d_2}{d_1 + d_2}$

$\frac{A(d_1 + d_2)}{d_1 + d_2}$

Q17. In the given figure, PQRS is a rectangle, a semicircle with SR as the diameter is drawn. A circle is drawn as shown in the figure given below. If QR = 7 cm, then what is the radius (in cm) of the small circle?



21 + 14√2

21 - 14√2

31 - 24√2

none of these

Q18. What is the value of $\log_9 27 - \log_{27} 9$?

2/3

4/7

5/6

6/7

Q19. The sum of the squares of the three consecutive even natural numbers is 1460. Find the numbers.

18, 20, 22

20, 22, 24

22, 24, 26

24, 26, 28

Q20. In an arithmetic progression, the first term is 10 and its common difference is 8. If the general term is a_n, find $a_{19} - a_{11}$.

65

66

64

70

Q21. **Fill in the blank with the appropriate preposition.**

The room is more expensive because it has a spectacular view _____ the beach.

for

from

of

to

Q22. In the question below, a statement is given followed by two courses of action numbered I and II. Assume everything in the statement to be true and on the basis of the information given in the statement, decide which of the suggested courses of action logically follow(s) for pursuing.
There is an unprecedented increase in the migration of villagers to urban areas as repeated crop failure has put them into a precarious financial situation.
Courses of Action:
I. The villagers should be provided with an alternate source of income in their villages which will make them stay put.
II. The migrated villagers should be provided with jobs in the urban areas to help them survive.

Only I follows

Only II follows

Neither I nor II follows

Both I and II follow

Q23. **The question consists of five statements labelled A, B, C, D, and E which when logically ordered form a coherent passage. Choose the option that represents the most logical order.**

A) People started fearing famine.
B) Monsoon turned out to be unusually abundant and the danger was averted.
C) The monsoon failed and water tanks became almost empty.
D) So, no grain could be sown by the farmers in their fields.
E) Farmers looked anxiously for the next monsoon.

CADBE

CDAEB

AEDCB

DABCE

Q24. Choose the appropriate option to complete the sentences.

You and _____ brother need to take time to prepare _____ for the long journey which will start next month.

his / yourself

yours / ourselves

their / you

your / yourselves

her / by themselves

Q25. **Fill in the blank:**

_____ we liked the technical support we received from the three companies, although X-Tech and Fast-on were a bit faster.

On the whole

After all

All in all

On the whole / All in all

Q26. Each question consists of four sentences on a topic. Some sentences are grammatically incorrect or inappropriate. Identify the incorrect sentence or sentences:

A. Chilika is situated on the eastern sea coast of India.
B. A narrow outer channel connects it to the Bay of Bengal.
C. On account of its rich biodiversity, Chitika was designated as a 'Ramsar site'.
D. Chilika supports some of the larger migratory birds in the country.

Only A

A and B

Only C

C and D

Q27. Choose the one that conveys the meaning of the sentence correctly.

It is too hard an essay for me to far attempt.

Too far it is an essay hard for me to attempt.

It is far too hard an essay for me to attempt.

It is too far hard an essay to attempt for me.

Q28. In the following question, a sentence has been given in Direct/Indirect speech. Out of the four alternatives suggested, select the one that best expresses the same sentence in Indirect/Direct speech.

He said to me, "You didn't help me in my financial crisis".

He told me that I did not helped him in his financial crisis.

He told me that I had not helped him in his financial crisis.

He told me that I have not helped him in his financial crisis.

He told me that I had not been helped him in his financial crisis.

Q29. **Convert the given sentence to active voice :**

He was obliged to resign by his boss

His boss obliged him to resign

To resign was his obligation to boss.

Circumstances obliged him to resign his boss.

Resignation obliged him to boss.

Q30. Identify the one that is opposite in meaning (antonym) to the question word and mark.

deface

disfigure

facelift

bungle

inter

Q31. Find out the synonym for the given word.

Purge

Thrust

Cognate

Erode

Evacuate

Q32. Choose an option, which can be substituted for a given sentence or phrase

A person who loves mankind and works for their well-being

Trustee

Philanthropist

Devotee

Disciple

Q33. **Find the missing term in the series:**

QPO, NML, KJI, _____, EDC

HGF

CAB

JKL

GHI

Q34. Pointing to a lady, Shahrukh said, “She is the daughter of the only child of my grandmother.” How is the lady related to Shahrukh?

Sister

Niece

Cousin

Data inadequate

Q35. Which day of the week was 27th October 1983?

Tuesday

Wednesday

Thursday

Friday

Q36. How many degrees will the minute hand move, at the same time at which the second hand move 4800?

80 deg

160 deg

140 deg

135 deg

Q37. In a certain code-language 'CUR WAP DIR' means 'red little box', 'SUT MAD BIX' means 'well arranged pile', 'BIX FAC DIR' means 'pile of boxes'. The code for 'of' here is

FAC

SUT

DIR

BIX

Q38. The question below consists of a question and two statements numbered A and B given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the question. Read both statements and give an answer.

What percent marks did Ram get in a test of 4 subjects?
(A) Ram got 90 in English and 84 in Maths.
(B) Ram got 75 in Hindi and 76 in Sanskrit.

if the question can be answered by using statement A alone but not by using B alone.

if the question can be answered by using either statement alone.

if the question can be answered by using both the statements together but not by either statement.

if both the statements taken together are not sufficient and more information is required to solve the question

Q39. **The question below consists of a question and two statements numbered A and B given below it. You have to decide whether the data provided in the statements are sufficient to answer the questions. Read both statements and give an answer.**

Five students Atul, Bala, Chetan, Dev and Ernesto were the only ones who participated in a quiz contest. They were ranked based on their scores in the contest. Dev got a higher rank compared to Ernesto, while Bala got a higher rank compared to Chetan. Chetan's rank was lower than the medium. Who among the five got the highest rank?

Statement A: Atul was the last rank holder.
Statement B: Bala was not among the top two rank holders.

If the question can be answered by using the statement A alone but not by using the statement B alone

If the question can be answered by using the statement B alone but not by using the statement A alone

If the question can be answered by using either of the statements alone

If the question can be answered by using both the statements together but not by either of the statements alone

Q40. The town of Paranda is located on Green Lake. The town of Akram is west of Paranda. Tokhanda is east of Akram but west of Paranda. Kokran is east of Bopri but west of Tokhanda and Akram. If they are all in the same district, which town is the farthest west?

Paranda

Kokran

Akram

Bopri

Q41. **Find the wrong term in the given number series.**
10, 17, 26, 37, 50, 65, 81

50

81

17

37

Q42. **Directions:** In the following question, only one Conclusion is given and five statements are given as a, b, c, d, and e. From this, you have to take the statements to be true even if they seem to be at variance with commonly known facts and then decide which of the given statements logically follows.

Conclusions :
Some gold which is white are not yellow.
Some gold being yellow is a possibility.
Some white are gold.
No silver is yellow.

a) Statements: No gold is silver. All silvers are white. No white is yellow.
b) Statements: Some gold is silver. Some silvers are white. No white is yellow
c) Statements: Some gold is not white. Some yellow is not white. No silver is yellow
d) Statements: Some gold is silver. All silvers are white. No white is a yellow
e) Statements: No gold is silver. No silvers are white. No white is a yellow

(a)

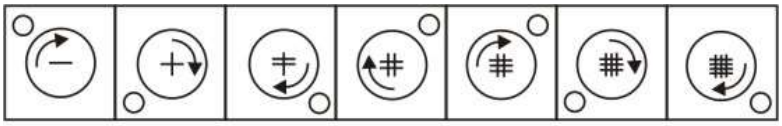
(b)

(c)

(d)

(e)

Q43. In the following question, the first and the last blocks mark the beginning and the end of a series respectively. The numbered blocks are a part of the series. However, one of these numbered blocks violates the pattern. Find the odd block and mark your answer from 1-5 accordingly.



1

2

3

4

Common Content:

Choose the appropriate words to complete the passage.

Decades ago, China ____1_____ the concept of barefoot doctors. They were community healthcare workers who successfully ____2___ the health of China's villages. Following this example, many African, Asian, and Latin American countries have started ____3___ programs. The largest of such community health efforts is India's National Rural Health Mission, in ___4____ over three years, the program has mobilized over fifty thousand new community health workers, each _____5____ as Asha.

Q44. Choose an appropriate word for the blank numbered (4)

course

approximate

period

just

Q45. Choose an appropriate word for the blank numbered (1)

researches

enlightened

introduced

originates

Q46. Choose an appropriate word for the blank numbered (2)

dealt

cured

entrusted

improved

Q47. Choose an appropriate word for the blank numbered (5)

preferred

perceived

known

regarded

Common Content:

Read the passage and answer the questions that follow.

One of the main reasons for corruption in elections today is the lure of power which haunts the politicians so much that they feel no qualms of conscience in adopting any underhand method to come out successful. The Watergate Scandal in the U.S.A. is an eloquent example to testify to the fact that even top-level politicians can stoop to the lowest level in order to maintain themselves in power. Who does not remember how Adolf Hitler rode roughshod overall canons of electoral propriety to capture power? In India also the record of the various political parties is not clean. Corruption thrives in elections because of those in the field playing on the psychology of the electorate. The voters are swayed by the tall promises of the candidates to whose machinations they fall easy prey. They are also susceptible to falling easy prey to the adulations of politicians due to their illiteracy. Besides, in the representative democracies, today and particularly in big countries the constituencies are quite extensive obviating the possibility of corrupt practices being discovered. Anti-corruption laws are honoured more in their breach than in their observance. Even the code of conduct to be observed by the parties fighting the elections becomes a dead letter in as much as it is jettisoned out of existence and thrown unscrupulously overboard by the unfair politicians whose only aim is to maintain themselves in the saddle.

Q48. Which one of the following may be the most appropriate title for the above passage?

- Corruption in Public Life
- Corruption in High Places
- Politicians Game of Power
- Elections and Corruption
- The True Character of a Politician

Q49. Adolf Hitler came to power :

- As a result of bungling in elections
- By liquidating any semblance of opposition
- By organising a mass movement
- With foreign help
- By conquest

Q50. Why according to the writer, do the corrupt practices indulged in elections go unnoticed?

- Because nobody is interested in discovering corrupt practices
- Because vested interests shield those who indulge in corrupt practices
- Because constituencies are so small that the politicians take the electorate, whose number is very limited, into confidence
- Because the law enforcing authorities themselves are in collusion with the politicians
- Because the constituencies are so big that it becomes difficult to discover the corrupt practices

Q51. How does corruption thrive in elections?

- The people themselves are corrupt
- A sizable part of the society is corrupt
- There is natural connection between elections and corruption

The politicians exploit the electorate psychologically

The politician cheat the electorate economically

Common Content:

These questions are based on the following information.
A cube, painted on all its faces, is cut into 125 identical smaller cubes.

Q52. How many smaller cubes have exactly two faces painted?

36

54

48

60

Common Content:

Read the given information carefully and answer the questions given beside:

12 friends are sitting in 2 parallel rows containing 6 persons each, in such a way that there is an equal distance between the adjacent persons. In row 1 P, Q, R, S, T, and U, and all of them are facing south. In row 2 J, K, L, M, N, and O are facing North. M is sitting 3rd to the left of J. N is facing R. P is sitting 3rd to the right of T. Q is facing J and sitting at one end of the row. U is sitting second to the right of T. K is facing S and he is not sitting in the middle position of the row. The person facing T is second to the right of M. L is not facing T and he is not sitting on the immediate left of J.

Q53. Who among the following is sitting second to the left of M?

O

J

N

K

Q54. Who is sitting second to the right of the person who is facing N?

Q

J

R

P

Q55. In the row facing south who is sitting at the ends of the row?

P, U

S, Q

S, T

None of these

Q56. Who are the immediate neighbour of N?

L, M

N, M

O, J

M, O

Common Content:

Read the data carefully and answer the following questions.

Average annual salary offered (in lakhs of Rupees)

	Institutes	Placements within the country						Placements abroad					
		2000	2001	2002	2003	2004	2005	2000	2001	2002	2003	2004	2005
1	ENRON	6.5	5.2	8	10.7	11.6	5.19	20.5	21.4	21.7	26.8	27	23.8
2	HORISTEN	2.9	3.5	2.4	3.5	1.7	4.33	34.22	33.2	30.9	38.87	40.7	35.8
3	CALLONA	4.9	3.1	6	11.2	9.1	3.85	25.44	25.4	18.2	29.82	30.9	25.1
4	JOSEPH	9.9	9.9	1.5	4.5	2.8	3.62	19.5	31.3	24.9	19.67	19.4	15
5	PREMIER	3	5.4	4.1	10.3	35	6.91	51.2	50.3	40.7	48.12	47	36.4
6	BRIDGESTONE	5.6	4	5.4	6.7	2.8	6.87	34.9	34.7	21.9	27.59	23.9	18.2
7	TENON	1.4	1.4	1.3	5.8	4.2	3.13	22.7	23.9	18.2	27.35	29	25.2
8	VERDE	6.5	6.1	8.9	13.8	11.9	5.98	31.4	30.3	30	36.38	38	26.1
9	KALCI	6	5.5	5.4	6.9	7.3	3.95	59.3	60.7	35.3	38.57	29.4	24
10	MALVIN	15.1	9.8	13.5	21.1	29.7	8.91	49.2	50.8	22.1	50.43	51.1	34.8
11	DC	8.4	8.3	7.2	13.7	15.2	7.7	45.5	61.7	36.7	48.86	52.5	33.2
12	WINSIN	10.3	6.8	6.8	11.6	12.7	4.95	45.7	44.7	33.1	38.16	37.6	31.2
13	ZKIN	8	4.5	12.7	14.5	9	4.87	46.2	57.8	28.3	40.98	40.6	35.8
14	STAR	6.8	6.2	12.4	18.8	28	9.88	54.2	54.9	50.6	59.98	56.7	50.9
15	HINDALCO	6.3	5.9	8.5	14.3	19	6.33	55.3	49.8	35.2	41.88	41.1	35.9
16	CAMCON	8.4	6.6	7.8	13.3	12.4	5.98	43.1	46	31.8	39.6	40.1	33.1
17	TORA	4.5	3.6	4	7.1	5.6	3.82	19.2	18.8	14.7	18.1	18.1	15.4
18	UKAI	5.1	4.4	6.5	6.3	5.5	3.66	13.6	11.6	12.4	13.12	13.7	12.4

Q57. How many institutes have an average annual salary offered was less than 10 lakh in the year, 2004 for placements within the country?

7

8

9

10

Q58. If we arrange the institutes according to the descending order of the average annual salary offered in 2003 (placements within the country) and according to the descending order of the average annual salary offered in 2003 (placements abroad), then which of the following institutes has the same rank in both the arrangements?

- HIDALCO
- ZKIN
- CAMCON
- None of these

Q59. If we arrange the institutes in alphabetical order, then what is the average annual salary offered of the seventh institute in the given period for placements abroad? (in Rs. lakh)

- 35.6
- 42.3
- 25.5
- 20.4

Q60. If we arrange the institutes according to the descending order of the average annual salary offered in 2003 (placements within the country) and according to the descending order of the average annual salary offered in 2003 (placements abroad), then which of the following institutes rank fourteenth in the two arrangements respectively?

- ENRON and UKAI
- BRIDGESTON and TENON
- KALCI and BRIDGESTON
- TORA and VERDE

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. What will be the output of the following pseudocode?

```
1  integer a, b, c
2  set a= -3, b = -2
3  if( b - 2 < b)
4  for (each c from 1 - 1 to 2 - 1)
5  a= a + 1
6  end for
```

```
7  end if
8  print a +b
```

- -4
- 3
- 4
- -3

Q2. Which one of the following gets generated by the given pseudocode for input n=5?

```
1  function(input n)
2  {
3    if (n < 2)
4     return n;
5   else return function (n-1) + function(n-2);
6  }
```

- 5
- 4
- 3
- 1

Q3. What will be the output of the following pseudocode?

```
1  Integer n, c
2  set n= 128, c= 4
3  while( n NOT EQUALS 0)
4     n= n / 10
5     c = c - 1
6  end while
7  print c
```

- 0
- -1
- 2
- 1

Q4. What will be the output of the following pseudocode?

```
1  Integer a, b, c
2  set a= 1, b = 2 , c= 1
```

```
3 a= c
4 c= a
5 b= a
6 print a + b + c
```

- 3
- 13
- 9
- 15

Q5. What will be the output of the following pseudocode?

```
1 Integer a, b, c
2 set a= 7, b= 2, c= 8
3 b = (b&b) + b
4 a= (5 & 11) + a
5 c= c
6 print a + b + c
```

- 30
- 20
- 15
- 22

Q6. What will be the output of the following pseudocode?

```
1 Integer a,b,c
2 Set a=2,b=4,c=6
3 b=(c^b)&b
4 b=(b+1)+c
5 b=(c+11)+b
6 c=(8+11)&a
7 c=(c^12)+a
8 Print a+b+c
```

- 46
- 50
- 42
- 54

Q7. What will be the output of the following pseudocode?

```
1 set p=1, q=1
2 Integer fun ( integer p, integer q)
3 if( q ^ 2 > p ^ 3)
4 p= q ^ 2
5 else
6 q = p & 2
7 end if
8 return p ^ q
9 end function fun ()
```

- 3
- 8
- 2
- 1

Q8. What will be the output of the following pseudocode?

```
1 integer j,m
2 set m=0
3 integer a[4]={10,11,3,2}
4 for (each j from m to m+3)
5 if (j> 0)
6 a[j]= a[j-1] + 2
7 end if
8 end for
9 m= a[0] + a[2] + a[1]
10 print m
```

- 52
- 29
- 41
- 36

Q9. What will be the output of the following pseudocode for **p = 3** and **q = 7**?

```
1 Integer function funn(Integer p, Integer q)
2   Integer r
3   set r=p
4   p=q
5   r=p
6   q=r
7   p=r
8   return p+q
9 End function funn ()
```

- 14
- 16

15

17

Q10. What will be the output of the following pseudocode for a=9,b=2?

```
1  Integer funn(Integer a, Integer b)
2    if (b<a)
3      return a+funn(b+1,a+1)+funn(b+2,a+2)
4    End if
5    return b
```

37

27

41

30

Q11. What will be the output of the following pseudocode for **a=3**, **b=1**, **c=7**?

```
1  Integer funn(Integer a, Integer b, Integer c)
2    c=(3^1)+a
3    b=3&c
4    b=(a+7)^b
5    if ((a^c^3)<(a^b^1)
6      b=a+c
7      c=(1+3)+a
8    End if
9    return a+b+c
```

16

18

15

17

Q12. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=5, q=6, r=10
3  q= (r+q)^q
4  if((r^3)<(5+r))
5    q=(11&10)+p
6    p=(q^1)+q
7  End if
8  Print p+q+r
```

58

54

52

65

Q13. What will be the output of the following pseudocode if **p = 3** and **q = 5**?

```
1  Integer fun(Integer p, Integer q)
2    if(q>p)
3     return fun(q-1, p)+q
4    End if
5    return p
```

8

7

9

6

Q14. What will be the output of the following pseudocode?

```
1  Integer a,b,c
2  Set a=3,b=2,c=7
3  for (each c from 5 to 9)
4     b=b+c
5  End for
6  a=(5+10)+a
7  c=(12+10)^c
8  for (each c from 5 to 6)
9     a=(c+b)+b
10    b=(a^9)+b
11 End for
12 Print a+b
```

536

540

547

541

Q15. What will be the output of the following pseudocode?

```
1  Integer p, q, r
2  Set q = 13
3  for (each p from 1 to 4)
4     r = q mod p
5     p = p + 5
```

```
6     q = p + r
7   end for
8   r = q / 5
9   Print q , r
```

- q = 2, r = 1
- q = 6, r = 1
- q = 7, r = 2
- q = 6, r = 4

Q16. Which of the following memory is non-volatile and may be written only once?

- RAM
- EE-PROM
- EPROM
- PROM

Q17. Scheduling is

- Allowing job to use the processor
- Performance considerations
- Quiet simple to implement, even on large main frames
- The same regardless of the purpose of the system

Q18. Which of the following device can be used to directly input printed text?

- OCR
- Mouse
- MIC
- Joystick

Q19. The memory which is programmed at the time it is manufactured is _____.

ROM

RAM

PROM

EPROM

Q20. Which of the following techniques provides a dedicated communication channel between two stations?

Switch network

Circuit switching

Packet switching

None of the mentioned options

Q21. In a database, related fields are grouped to form

Record

File

Bank

Field group

Q22. Which of the following is a serious problem of file management systems?

Data redundancy

Difficult to update

Program dependence

All of the mentioned options

Q23. A kilobyte also referred to as KB, is equal to

10000 bytes

1024 bytes

- 2048 bytes
- 512 bytes

Q24. The process of putting data into a storage location is called

- Reading
- Writing
- Controlling
- Hand shaking

Q25. A gate in which all inputs must be low to get a high output is called

- An inverter
- A NOR gate
- An AND gate
- A NAND gate

Q26. A collection of wires that connects several devices is called

- Link
- Bus
- Cable
- Port

Q27. A source program is a program written in _____ language.

- English
- Symbolic
- High-level
- Object

Q28. Which of the following is the ascending order of the data hierarchy?

- bit - byte - record - field - database - file
- bit - byte - field - record - file - database
- bit - byte - file - field - record - database
- bit - record - byte - field - file - database

Q29. Which of the following is the product of data processing?

- Information
- Data
- Software program
- System

Q30. Multiprogramming systems

- Are easier to develop than single programming systems
- Execute each job faster
- Execute more jobs in the same time period
- Use only one large mainframe computer

Answer Key & Solution

Section 1 - Aptitude

Q1

33

Solution

135/5 = 27

27/5 = 5

5/5 = 1

therefore, 27+5+1 = 33

Q2

p = 2, q = −7

Solution

Let $f(x) = px^3 - x^2 + qx + 6$.

Since it is divisible by x+2,

we have f(-2) = -8p -4 -2q+6=0

=> 4p+q-1=0…… (i)

Similarly, since it is divisible by x-3, we have

f(3) = 27p-9+3q+6=30

f(3) = 9p+q-11=0…….(ii)

Solving, we get p=2, q=-7.

Q3

800

Solution

Let the number be x.

(75/100)x - (65/100)x = 80

> x= 800 .·.

Twice of 800 = 1600.

Q4

168

Solution

The required number of selections = ${}^7C_1 \times {}^6C_1 \times {}^4C_1 = 168$.

Q5

44

Solution

Applying the derangement formula, the number of ways in which this error could have been done is given by:

$D_5 = 5! \times (1/2! - 1/3! + 1/4! - 1/5!)$

$= 120 \times (1/2 - 1/6 + 1/24 - 1/120)$

$= 44$

The teacher could have made this error in 44 different ways.

Q6

Rs. 30000

Solution

723 more than the compound interest for 2 yrs at 20% half yearly than annually

$P(1 + 20/200)^4 - P - P(1 + 20/100)^2 + P = 723$

$14641P/10000 - 14400P/10000 = 723$

$241P/10000 = 723$

$P = 30000$

Q7

30

Solution

Simple interest=PNR/100 = (12000*10*1)/100 = 1200

For C.I, since it is reckoned half yearly, N = 2 years and R = 5%

compound interest $=(12000*(1+(5/100))^2-12000) = 1230$

Difference = 1230 - 1200 = 30

Q8

6 days

Solution

Adams, Biden, Carter and Dwight will together take $1/8 + 1/24 = 1/6 \Rightarrow$ 6 days to complete the work

Q9

230 m

Solution

Relative speed = (120 + 80) km/hr

[200 x 5/18] m/sec = [500/9] m/sec.

Let the length of the other tain be x metres.

Then, (x + 270) / 9 = 500/9

⇔ x + 270 = 500

⇔ x = 230 m

Q10 28/25

Solution

1st case - 12 litre milk in 15 (12+3) litre content

.Ratio = 12/15 = 4/5

2nd case - 10 litre milk in 14 (10+4) litre content.

Ratio = 10/14 = 5/7

Ratio of milk strength in two = 4/5 : 5/7

= 28 : 25

Q11 1 : 2

Solution

	Milk	:	Water
Vessel A	3	:	2
Vessel B	7	:	3

Now using alligation,

Vessel 1	:	Vessel 2
3/5		7/10
	2/3	
(7/10-2/3)=1/30		(2/3-3/5)=1/15
1	:	2

Q12 2/25

Solution

Favourable ways (29, 92, 38, 83, 47, 74, 56, 65)

Hence required 8/100

2/25

Q13

40000

Solution

let the price of fan be X and fridge be Y

6x/100+9y/100 = 5100

6x+9y=510000------(1)

then 9x/100-6y/100=1800

9x-6y=180000-------(2)

Calculate both, the we will get

x=Rs40000

Q14

Rs 14160

Solution

80% of M.P = 9600

∴ M. P. = Rs 12000

S.P. for second person = $12000 \times \frac{118}{100}$

= Rs. 14160

Q15

32

Solution

Let the numbers be a+b+c+d+e+f+g+h= 19*8= 152

a+f= 34

c+e+g= 72

=> b+d+h= 152-(34+72)= 46

d-b=8

b-h=4

Solving the equations,

b= 14, d= 22, h= 10

Sum of 4th and 8th number= 22+10= 32

Q16 $\frac{2A(d_1 + d_2)}{d_1 d_2}$

Solution

AB . d_1 = A and AD . d_2 = A

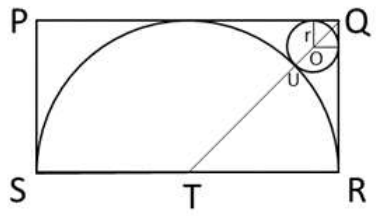
AB=A/d_1 and AD=A/d_2

The perimeter of the parallelogram = 2 (AB + AD) =2(A/d1 + A/d2)

Ans(1)

Q17 21 - 14√2

Solution



As we can see in the figure, RT and QR are the radius of the bigger circle and equal to 7 cm,

⇒ QT^2 = 49 + 49

⇒ QT = 7√2

Now suppose the radius of the smaller circle is r cm,⇒ QO = r√2

Since QT = TU + UO + OQ ⇒ QT = 7 + r + r√2

⇒ 7√2 = 7 + r(√2 + 1)

⇒ r = [7(√2 - 1)/(√2 + 1)]

⇒ r = 7(3 - 2√2)

⇒ r = 21 - 14√2

Q18 5/6

Solution

$\log_9 27 = m \Leftrightarrow 9^m = 27 \Leftrightarrow 3^{2m} = 3^3 \Leftrightarrow 2m = 3.$

⇒ m = 3/2

And, $\log_{27} 9 = n \Leftrightarrow (27)^n = 9 \Leftrightarrow 3^{3n} = 3^2 \Leftrightarrow 3n = 2$

⇒ n = 2/3

∴ $\log_9 27 - \log_{27} 9 = 3/2 - 2/3 = (9-4)/6$

⇒ 5/6.

Q19

20, 22, 24

Solution

Three consecutive even natural numbers be 2x - 2, 2x and 2x + 2.

$(2x - 2)^2 + (2x)^2 + (2x + 2)^2 = 1460$

$4x^2 - 8x + 4 + 4x^2 + 4x^2 + 8x + 4 = 1460$

$12x^2 = 1452 \Rightarrow x^2 = 121 \Rightarrow x = \pm 11$

As the numbers are positive, 2x > 0. Hence x > 0. Hence x = 11.

Required numbers are 20, 22, 24.

Q20

64

Solution

Given a = 10 and common difference = 8.

General Term $a_n = a +(n-1)d \Rightarrow a_n = 10 + 8(n-1) = 8n + 2$

$a_{19} = 152 + 2=154$

$a_{11} = 88 + 2 = 90$

$a_{19} - a_{11} = 64$

Q21

of

Solution

The meaning of the given sentence "a window that offers a nice image for guests"

The noun **view should be used in as *view (of) something / someplace***

Hence the appropriate preposition is **of**

Usage of From : indicating the point at which an observer is placed.

The observer is watching the beach from the room and not vise-versa Hence it is incorrect.

Q22

Only I follows

Solution

Clearly, increased migration would add to the burden on city's infrastructure. So, attempts should be made to make the villagers feel comfortable in the villages itself. So, only course I follows.

Q23

CDAEB

Solution

The paragraph clearly identifies 3 ways:

The Para starts with a failing monsoon and because of this tanks became empty.

This fear led to no sowing of the grains by the farmers.

This led to fear of famine

After this farmers again became anxious for the next monsoon.

After this speculation, something opposite happened the monsoon fared well and danger was averted. The correct sequence is CDAEB

Q24

your / yourselves

Solution

The pronoun here should denote the brother belonging to him , thus the appropriate pronoun would be "your"

Q25

After all

Solution

After all - in spite of any indications or expectations to the contrary.

Q26

C and D

Solution

Sentence C should begin with 'On account of ....' (meaning because of) and not with 'on an account of ...'

In sentence D it should be '... the largest of migratory birds,' meaning large in number. As given it suggests large in size.

Sentences A and B are grammatically correct.

Q27

It is far too hard an essay for me to attempt.

Solution

It is far too hard an essay for me to attempt is correct because far should come before 'too'

Q28

He told me that I had not helped him in his financial crisis.

Solution

Here the reporting verb(said) is in Past tense.

The Simple Past tense(didn't help) in the reported speech will be changed to Past Perfect tense(had not helped) in Indirect form.

Also, pronoun ‘you’ will be replaced by ‘I’, ‘me’ by ‘him’ and ‘my’ by ‘his’ in the indirect form.

Therefore, the correct indirect form is He told me that I had not helped him in his financial crisis.

Q29

His boss obliged him to resign

Solution

The given sentence is in Passive Voice."boss" is object. "He" is subject. So when there is a change of voice,the boss becomes subject and" he" changes to "him". The right answer is "His boss obliged him to resign" All the other options make no sense.

Q30

facelift

Solution

'Deface' is to spoil the appearance of something. 'Bungle' means ruin. Hence, out of the given options 'facelift' is the antonym of 'deface'.

Q31

Evacuate

Solution

Explanation:

Purge means the act of removing people or things that are not wanted

Thrust - push suddenly or violently in a specified direction.

Cognate - related; connected.

Erode - (of wind, water, or other natural agents) gradually wear away (soil, rock, or land).

Evacuate - remove (someone) from a place of danger to a safer place.

Q32

Philanthropist

Solution

Trustee - an individual person or member of a board given control or powers of administration

Philanthropist - a person who seeks to promote the welfare of others

Devotee - a person who is very interested in and enthusiastic about someone or something

Disciple - a personal follower

Q33

HGF

Solution

This series consists of letters in a reverse alphabetical order.

Q34 Sister

Solution

Only child of Shahrukh's grandmother — Shahrukh's father/mother.

Daughter of Shahrukh's father/mother — Shahrukh's sister.

Q35 Thursday

Solution

27th October 1983 => 1982 completed years.

1982 can be split as 1600 + 300 + 82

1600 years => 0 odd days;

300 years = 1 odd day and 82 years => 20 leap years and 62 non-leap years.

20 leap years consists of 40 or 20 × 2 = 40 odd days or 5 odd days.

62 non-leap years consists of 62 × 1 = 62 odd days or 6 odd days.

∴ A total of 0 + 1 + 5 + 6 = 5 odd days in 1982 years.

In the year 1983,

Month : Jan Feb Mar Apr May Jun

No of days : 31 28 31 30 31 30

Odd days : 3 + 0 + 3 + 2 + 3 + 2 = 13 odd days = 6 odd days

Month : Jul Aug Sep Oct

No of days : 31 31 30 27

Odd days : 3 + 3 + 2 + 6 = 0 odd days

5 + 6 = 4 odd days

Hence, 27th October 1983 is Thursday.

Q36 80 deg

Solution

As minute hand covers, 60 degrees

Minute hand covers 4800/60 = 80°

Q37 FAC

Solution

CUR WAP DIR --------(1)

red little boxes

BIX FAC DIR -------(2)

pile of boxes.

So DIR = boxes

BIX FAC DIR -------- (2)

pile of boxes

SUT MAD BIX --------- ---(3)

well arrange pile

So BIX = pile

Thus of = FAC

Q38 if both the statements taken together are not sufficient and more information is required to solve the question

Solution

From both the statements, we get the total marks of Ram but nothing is mentioned about the maximum marks of any subject. So we cannot answer the question.

Q39 If the question can be answered by using the statement A alone but not by using the statement B alone

Solution

By using the statement (A)alone we get the ranking of all the five students as below

Dev	Ernesto	Bala	Chetan	Atul
1	2	3	4	5

Q40 Bopri

Solution



Bopri is the farthest West.

Q41 81

Solution

10 + 7 = 17

17 + 9 = 26

26 + 11 = 37

37 + 13 = 50

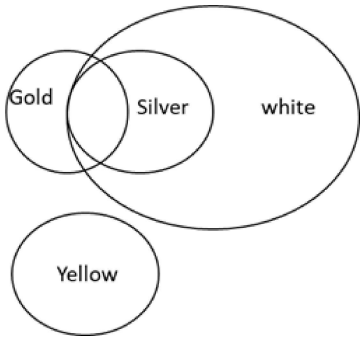
50 + 15 = 65

65 + 17 = 82

Hence, in place of 81, it should be 82.

Q42 (d)

Solution



Statements (d) satisfies the given conclusions.

Q43 4

Solution

The arrow rotates clockwise; the vertical and horizontal lines are added alternately. The smaller circle moves from corner to corner in an anticlockwise direction. In figure (4), the smaller circle should be in the top left corner.

Ans.(4)

Q44. just

Solution

No Solution

Q45. introduced

Solution

Since the passage talks about something that happened decades ago , the verb should be in the past tense. Hence the option originates is eliminated.

The only other option that fits in the context of the passage is introduced.

Q46. improved

Solution

The health of the villages improved , the appropriate word for the blank.

Q47. known

Solution

The passage talks about the implementation of the idea in India, which is known as Asha .Hence the answer.

Q48. Elections and Corruption

Solution

The passage on the whole talks only about the politicians and their politics

Q49. As a result of bungling in elections

Solution

Third line of the paragraph says"Who does not remember how Adolf Hitler rode roughshod overall canons of electoral pro-priety to capture power ?"

roughshod- shoes with nail

The sentence says his inappropriate behavior towards the electoral rules

Bugling means clumsy and not appropriate

Q50. Because the constituencies are so big that it becomes difficult to discover the corrupt practices

Solution

The last three lines of the passage clearly explains this

Q51. The politicians exploit the electorate psychologically

Solution

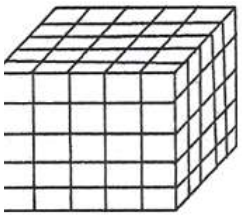
Fifth line of the para clearly says that people in politics play with the psychology of those in electoral

Q52. 36

Solution

Here $n^3 = 125$

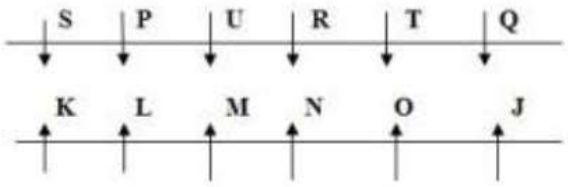
$\Rightarrow n = 5$



ubes with two faced painted: 12(n – 2) = 12(3) = 36.

Q53. K

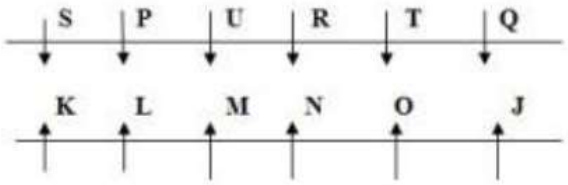
Solution



K is sitting second to the left of M

Q54. P

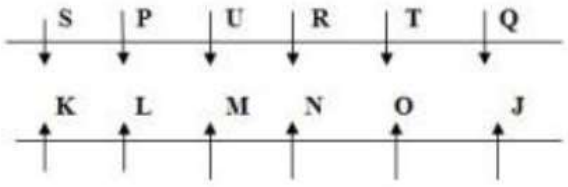
Solution



P is sitting second to the right of the person who is facing N.

Q55. S, Q

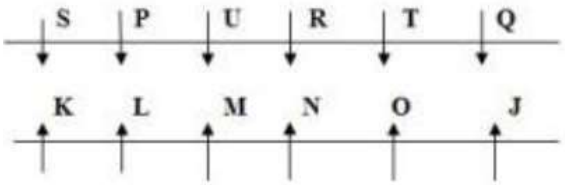
Solution



S and Q are sitting at the ends facing South.

Q56. M, O

Solution



I and O are the immediate neighbours of N.

Q57. 9

Solution

Institute, which have average annual salary offered was less than 10 lakhs in the year 2004 for placement within the country are :

1.HORISTEN 2. CALLONA 3.JOSEPH 4. BRIDGESTONE 5. TENON 6. KALCI 7. ZKIN 8. TORA 9. UKAI

Q58. CAMCON

Solution

Institutes	2003	Rank	2003	Rank
1.ENRON	10.7	10	26.8	15
2.HORISTEN	3.5	18	38.87	8
3.CALLONA	11.2	9	29.82	12
4.JOSEPH	4.5	17	19.67	16
5.PREMIER	10.3	11	48.12	4
6.BRIDGESTONE	6.7	14	27.59	13
7.TENON	5.8	16	27.35	14
8.VERDE	13.8	5	36.38	11
9.KALCI	6.9	13	38.57	9
10.MALVIN	21.1	1	50.43	2
11.DC	13.7	6	48.86	3
12.WINSIN	11.6	8	38.16	10
13.ZKIN	14.5	3	40.98	6
14.STAR	18.8	2	59.98	1
15.HINDALCO	14.3	4	41.88	5
16.CAMCON	13.3	7	39.6	7
17.TORA	7.1	12	18.1	17
18.UKAI	6.3	15	13.12	18

From the table we can find that after arranging in descending order we will get same Rank for CAMCON which is at 7th rank both way.

Q59. 35.6

Solution

Arrangement of the 7 institutes in alphabetical order

1. BRIDGESTONE 2.CALLONA 3. CAMCON 4. DC 5. ENRON 6. HINDALCO 7. HORISTENE

7th institute in alphabetical order is HORISTENE and average annual salary offered by this institute for placement abroad = (34.22+33.2+30.9+38.87+40.7+35.8)/6 = 35.6 lakhs

Q60. BRIDGESTON and TENON

lution

y arranging the institutes based on the average salary offered in descending order, we can

nd that BRIDGESTON and TENON rank fourteenth in arrangement.

Section 2 - Pseudocode and Computer Fundamentals

Q1

-3

Solution

Initially a=-3, b= -2.

For loop gets executed twice and increment the 'a' value (i.e) a= -1 when loops terminates.

As a result , -1 +(-2) = -3 (gets printed)

Q2

5

Solution

Given pseudocode is used to find fibonacci series. So fibonacci series of 5 is 5.

Q3

1

Solution

The logic of the program is to find a number of digits in the number. The total digit for the given number 128 is 3, so 'c' decrements three times (4-3 = 1)

As a result, 1 gets printed.

Q4

3

Solution

Value of 'c' copied to 'a', then 'a' value copied to 'b' and 'c'.

As a result, the value of c is assigned to all variables.

Output: 1+1+1 =3.

Q5

20

Solution

Fibonacci Sequence:

The Fibonacci sequence is defined as:

- `F(0) = 0`
- `F(1) = 1`

- For `n >= 2`, `F(n) = F(n-1) + F(n-2)`

Calculating Fibonacci Number for `n = 5`:

1. `function(5)` calls `function(4)` and `function(3)`.
2. `function(4)` calls `function(3)` and `function(2)`.
3. `function(3)` calls `function(2)` and `function(1)`.
4. `function(2)` calls `function(1)` and `function(0)`.
5. `function(1)` returns `1`.
6. `function(0)` returns `0`.

Now, let's calculate:

- `function(2) = function(1) + function(0) = 1 + 0 = 1`
- `function(3) = function(2) + function(1) = 1 + 1 = 2`
- `function(4) = function(3) + function(2) = 2 + 1 = 3`
- `function(5) = function(4) + function(3) = 3 + 2 = 5`

So, the output of the pseudocode for input `n = 5` is `5`.

Q6

42

Solution

No Solution

Q7

2

Solution

No Solution

Q8

36

Solution

Initially m=0, a[4]={10,11,3,2}.

The current index array value is assigned inside the for loop by adding the previous array value by two.

End of the loop a[4]= {10 12 14 16}

m=10+12+14 => 36 gets printed.

Q9

14

Solution

No Solution

Q10

30

Solution

In funn(9,2) = 2<9(True) = 9+funn(3,10)+funn(4,11).

In funn(3,10) = 10<3 (False) = return 10 to funn(3,10)

In funn(4,11) = 11<4 (False) = return 11 to funn(3,10).

So 9+10+11= 30.

Q11

18

Solution

No Solution

Q12

54

Solution

Initially p=5,q=6 and r=10. If block gets executed and update p=29,q=15. So output is 29+15=54.

Q13

9

Solution

No Solution

Q14

540

Solution

In for loop b=2+3=>5 ,same process repeated till c=9 (finally b=37). a=5+10+3=18. c=(12+10)^c=31

In next loop, when c=5, a=5+37+37 => 79 and b=79^9+37=107. when c=6, a=6+107+107 => 220 and b=320.

So 220+320 =>540

Q15

q = 6, r = 1

Solution

Initially q=13, Loop starts from 1 i.e p=1.

r= 13%1=> 0

p=1+5=>6

q=6+0 =>6

Here p> 4 so loop terminates.

r=6/5=>1

Therefore 6 1 gets printed.

Q16 PROM

Solution

A programmable read-only memory (PROM) or field programmable read-only memory (FPROM) or one-time programmable non-volatile memory (OTP NVM) is a form of digital memory where the setting of each bit is locked by a fuse or anti-fuse. It is one type of ROM (read-only memory). The data in them is permanent and cannot be changed.

Q17 Allowing job to use the processor

Solution

No Solution

Q18 OCR

Solution

Optical character recognition (also optical character reader, OCR) is the mechanical or electronic conversion of images of typed, handwritten, or printed text into machine-encoded text, whether from a scanned document, a photo of a document, a scene-photo or from subtitle text superimposed on an image.

Q19 ROM

Solution

ROM is an acronym for Read-Only Memory. It refers to computer memory chips containing permanent or semi-permanent data. Unlike RAM, ROM is non-volatile; even after you turn off your computer, the contents of ROM will remain. Almost every computer comes with a small amount of ROM containing the boot firmware.

Q20 Circuit switching

Solution

No Solution

Q21 Record

Solution

No Solution

Q22 All of the mentioned options

Solution

No Solution

Q23

1024 bytes

Solution

1 KB = 1024 bytes.

Q24

Writing

Solution

No Solution

Q25

A NOR gate

Solution

No Solution

Q26

Bus

Solution

No Solution

Q27

High-level

Solution

No Solution

Q28

bit - byte - field - record - file - database

Solution

No Solution

Q29

Information

Solution

No Solution

Q30

Execute more jobs in the same time period

Solution

No Solution